

Why Variational Mean Field Games Are Convex

Momentum variables, perspective functions, and the Benamou–Brenier structure

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Abstract

A basic variational mean field game minimizes kinetic effort and congestion over a population curve whose initial distribution is prescribed and whose final distribution is penalized. In density and velocity variables, neither the kinetic energy $\rho|v|^2/2$ nor the constraint $\partial_t \rho + \operatorname{div}(\rho v) = 0$ is jointly convex. Convexity appears after introducing the mass flux $m = \rho v$: the constraint becomes affine and the kinetic energy becomes $|m|^2/(2\rho)$, the perspective of a quadratic function. This note proves that mechanism carefully, including the extension at vacuum, the hypotheses on the congestion potential, the convex dual, and the Hamilton–Jacobi/continuity optimality system. It also separates this full-horizon problem from Benamou–Brenier transport and JKO minimizing movements.

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1 The model and the short answer

Let $\mathcal{X} = \mathbb{T}^d$, or let $\mathcal{X} = \mathbb{R}^d$ under suitable moment assumptions. The initial probability measure is $\alpha_0 = \rho_0 dx$. A population curve $\alpha_t = \rho_t dx$ is transported by a velocity v_t according to

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0, \quad \rho_{t=0} = \rho_0. \quad (1)$$

The final density is free and is charged through a terminal potential $\Psi : \mathcal{X} \rightarrow \mathbb{R}$. Given a congestion potential $G : [0, +\infty) \rightarrow \mathbb{R} \cup \{+\infty\}$, consider

$$\inf_{\rho, v} \left\{ \int_0^1 \int_{\mathcal{X}} \left(\frac{1}{2} \rho_t |v_t|^2 + G(\rho_t) \right) dx dt + \int_{\mathcal{X}} \Psi \rho_1 dx \right\} \quad (2)$$

subject to (1) and $\rho \geq 0$. This is a standard variational MFG model (Benamou and Carlier, 2015; Benamou et al., 2017). The factor 1/2 is only a normalization.

Definition 1.1 (Momentum or mass flux). The momentum variable is

$$m_t(x) = \rho_t(x)v_t(x). \quad (3)$$

Thus $m_t dx$ is a vector-valued measure recording transported mass per unit time, rather than velocity per particle.

In the variables (ρ, m) , the problem becomes

$$\inf_{\rho, m} \left\{ \int_0^1 \int_{\mathcal{X}} \left(\frac{|m_t|^2}{2\rho_t} + G(\rho_t) \right) dx dt + \int_{\mathcal{X}} \Psi \rho_1 dx \right\}, \quad (4)$$

subject to the affine conservation law

$$\partial_t \rho_t + \operatorname{div} m_t = 0, \quad \rho_{t=0} = \rho_0. \quad (5)$$

The quotient must be extended correctly at $\rho = 0$, as explained below.

Definition 1.2 (Convex variational MFG). Problem (4) is a convex optimization problem provided that G is convex: the kinetic integrand is jointly convex in (ρ, m) , the congestion is convex in ρ , the terminal cost is linear, and the constraints are affine.

2 Why the velocity formulation is not convex

The product $\rho|v|^2$ is convex in each variable separately, but separate convexity does not imply joint convexity.

Proposition 2.1 (Nonconvexity in density–velocity variables). *For $v \neq 0$, the map*

$$(\rho, v) \mapsto \frac{1}{2} \rho |v|^2, \quad \rho > 0,$$

is not jointly convex. Moreover, $\partial_t \rho + \operatorname{div}(\rho v) = 0$ is nonlinear in (ρ, v) .

Proof. In one dimension, the Hessian of $K(\rho, v) = \rho v^2/2$ is

$$\nabla^2 K(\rho, v) = \begin{pmatrix} 0 & v \\ v & \rho \end{pmatrix},$$

whose determinant is $-v^2 < 0$. It is therefore indefinite whenever $v \neq 0$. The multidimensional claim follows by restriction to a line. The product ρv in the continuity equation is bilinear. $\square$

The nonlinear change $(\rho, v) \mapsto (\rho, m = \rho v)$ does two things at once: it linearizes conservation and changes the kinetic integrand. At vacuum, v is physically irrelevant, while finite action forces $m = 0$.

3 The perspective mechanism

Definition 3.1 (Closed perspective). Let $L : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, convex, and lower semicontinuous. Its closed perspective is

$$J_L(a, m) = \begin{cases} aL(m/a), & a > 0, \\ L^\infty(m), & a = 0, \\ +\infty, & a < 0, \end{cases} \quad (6)$$

where L^∞ is the recession function of L .

For the quadratic Lagrangian $L(v) = |v|^2/2$, this is

$$J(a, m) = \begin{cases} \frac{|m|^2}{2a}, & a > 0, \\ 0, & (a, m) = (0, 0), \\ +\infty, & a = 0, m \neq 0, \\ +\infty, & a < 0. \end{cases} \quad (7)$$

This boundary convention is essential; the symbol $0/0$ is not a definition.

Proposition 3.2 (Convexity of the perspective). *If L is proper, convex, and lower semicontinuous, then J_L is jointly convex, lower semicontinuous, and positively one-homogeneous:*

$$J_L(\lambda a, \lambda m) = \lambda J_L(a, m) \quad (\lambda \geq 0).$$

Proof. Take $a_0, a_1 > 0$, $m_0, m_1 \in \mathbb{R}^d$, and $\theta \in [0, 1]$. Set

$$a_\theta = (1 - \theta)a_0 + \theta a_1, \quad \lambda_0 = \frac{(1 - \theta)a_0}{a_\theta}, \quad \lambda_1 = \frac{\theta a_1}{a_\theta}.$$

Then $\lambda_0 + \lambda_1 = 1$ and

$$\frac{(1 - \theta)m_0 + \theta m_1}{a_\theta} = \lambda_0 \frac{m_0}{a_0} + \lambda_1 \frac{m_1}{a_1}.$$

Convexity of L , multiplied by a_θ , proves the convexity inequality. The recession value at $a = 0$ is precisely the lower-semicontinuous closure. Homogeneity follows from the definition. $\square$

For the quadratic case, the epigraph is the rotated second-order cone

$$\text{epi } J = \{(a, m, s) : a \geq 0, s \geq 0, |m|^2 \leq 2as\},$$

which gives another direct proof of convexity.

Remark 3.3 (A recurring convexity mechanism). The same perspective transform makes $b\varphi(a/b)$ jointly convex in the theory of φ -divergences. Here it turns a per-particle velocity cost into a convex density–flux cost.

4 The convex primal problem

Define

$$\mathcal{E}(\rho, m) = \int_0^1 \int_X (J(\rho_t, m_t) + G(\rho_t)) \, dx \, dt + \int_X \Psi \rho_1 \, dx. \quad (8)$$

The variational MFG is

$$\inf \{ \mathcal{E}(\rho, m) : \partial_t \rho + \text{div } m = 0, \rho_{t=0} = \rho_0 \}. \quad (9)$$

Theorem 4.1 (Convexity of the variational MFG). *If $G : [0, +\infty) \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, convex, and lower semicontinuous, then (9) is a convex optimization problem in (ρ, m) .*

Proof. The perspective J is jointly convex by Proposition 3.2; G is convex by assumption; integration preserves convexity; and the endpoint term is linear. The continuity equation and initial condition are affine constraints. $\square$

If G is nonconvex, the momentum substitution still convexifies the kinetic term and linearizes the PDE, but it cannot convexify G . A convex terminal functional $\mathcal{G}(\rho_1 dx)$ may replace the linear terminal term without changing the theorem. A fixed endpoint is also a convex affine constraint.

Proposition 4.2 (Basic existence). *Let $\mathcal{X} = \mathbb{T}^d$, let Ψ be lower semicontinuous and bounded below, and assume G is convex, lower semicontinuous, and superlinear:*

$$G(r)/r \longrightarrow +\infty \quad (r \rightarrow +\infty).$$

If the infimum is finite, then the lower-semicontinuous relaxation of (9) to density–flux measures admits a minimizer.

Proof. For a minimizing sequence, superlinearity gives uniform integrability of the densities. The kinetic action controls the Wasserstein metric variation of the curves. The continuity equation is closed under weak convergence of density–flux pairs. Convexity, lower semicontinuity, and one-homogeneity of J give lower semicontinuity of its integral, while the congestion term is lower semicontinuous by convexity and uniform integrability. Since Ψ is lower semicontinuous and bounded below, the terminal term is lower semicontinuous under narrow convergence. The direct method produces a minimizer. On $\mathbb{R}^d$, an additional moment or coercivity bound is needed to prevent escape to infinity. $\square$

Strict convexity. The perspective J is one-homogeneous and hence not strictly convex along rays. If G is strictly convex, any two minimizers have the same density. Once $\rho > 0$ is fixed, strict convexity of $m \mapsto |m|^2/(2\rho)$ identifies the flux.

5 Optimality conditions and the MFG PDE

The congestion potential G is not itself the pointwise cost perceived by one infinitesimal agent. The coupling appearing in the HJB equation is

$$g(r) = G'(r)$$

when G is differentiable, or a subgradient $g \in \partial G(r)$ in the nonsmooth case.

Theorem 5.1 (Euler–Lagrange system). *Assume a minimizer (ρ, m) is smooth, $\rho > 0$, and G is differentiable. Then there is a potential u such that*

$$\begin{cases} -\partial_t u + \frac{1}{2} |\nabla u|^2 = G'(\rho), & u_{t=1} = \Psi, \\ \partial_t \rho - \operatorname{div}(\rho \nabla u) = 0, & \rho_{t=0} = \rho_0, \\ m = -\rho \nabla u. \end{cases} \quad (10)$$

This is the first-order quadratic MFG system with local coupling $g(\rho) = G'(\rho)$.

Proof. Introduce the multiplier $-u$ for the continuity equation. Integration by parts gives the Lagrangian

$$\begin{aligned}\mathcal{L}(\rho, m, u) = & \int_0^1 \int_X \left(\frac{|m|^2}{2\rho} + G(\rho) + \rho \partial_t u + m \cdot \nabla u \right) dx dt \\ & + \int_X (\Psi - u_1) \rho_1 dx + \int_X u_0 \rho_0 dx.\end{aligned}$$

Stationarity in m gives $m/\rho + \nabla u = 0$, hence $m = -\rho \nabla u$. Stationarity in ρ gives

$$-\frac{|m|^2}{2\rho^2} + G'(\rho) + \partial_t u = 0.$$

Substituting the flux relation yields the HJB equation. Since the terminal density is free and positive, terminal stationarity gives $u_1 = \Psi$. Primal feasibility gives the continuity equation. $\square$

Remark 5.2 (Vacuum and terminal complementarity). Without positivity, the terminal condition is

$$u_1 \leq \Psi, \quad (\Psi - u_1) \rho_1 = 0.$$

Thus equality holds on the support of ρ_1 . Likewise, $G'(\rho)$ is replaced by a subgradient when G is nonsmooth.

General kinetic costs. For a convex Lagrangian $L(v)$, use $J_L(\rho, m) = \rho L(m/\rho)$. With

$$H(p) = \sup_{v \in \mathbb{R}^d} \{-v \cdot p - L(v)\},$$

the optimality system becomes

$$-\partial_t u + H(\nabla u) = G'(\rho), \quad \partial_t \rho + \operatorname{div}(-\rho \nabla_p H(\nabla u)) = 0. \quad (11)$$

The quadratic case has $H(p) = |p|^2/2$.

6 The convex dual

Extend G by $+\infty$ on negative arguments and define

$$G^*(s) = \sup_{r \geq 0} \{rs - G(r)\}.$$

The restriction $r \geq 0$ implies that G^* is nondecreasing.

Proposition 6.1 (Dual problem). *Under a standard qualification condition, the dual of (9) is*

$$\sup_{u: u_1 \leq \Psi} \left\{ \int_X u_0 \rho_0 dx - \int_0^1 \int_X G^* \left(-\partial_t u + \frac{1}{2} |\nabla u|^2 \right) dx dt \right\}. \quad (12)$$

Proof. Use the integrated Lagrangian from Theorem 5.1. Pointwise in (t, x) ,

$$\inf_m \left\{ \frac{|m|^2}{2\rho} + m \cdot \nabla u \right\} = -\frac{1}{2} \rho |\nabla u|^2.$$

The remaining density infimum is

$$\inf_{\rho \geq 0} \left\{ G(\rho) + \rho \left(\partial_t u - \frac{1}{2} |\nabla u|^2 \right) \right\} = -G^* \left(-\partial_t u + \frac{1}{2} |\nabla u|^2 \right).$$

The infimum over the free terminal density is finite exactly when $u_1 \leq \Psi$. The fixed initial density contributes $\int u_0 \rho_0$, proving the formula. $\square$

At smooth primal–dual optimality, Fenchel equality gives

$$-\partial_t u + \frac{1}{2} |\nabla u|^2 = G'(\rho),$$

which is the HJB equation. The dual is indeed concave: the argument of G^* is convex in u , and G^* is convex and nondecreasing.

7 Examples of congestion

Example 7.1 (Power congestion). For $q > 1$,

$$G(r) = \frac{r^q}{q}, \quad g(r) = G'(r) = r^{q-1}.$$

This is strictly convex and superlinear. It penalizes concentration and gives the compactness used in Proposition 4.2.

Example 7.2 (Entropy congestion). Let $G(r) = r \log r$ for $r > 0$ and $G(0) = 0$. Then $g(r) = 1 + \log r$. Additive constants in the coupling only shift the value function, so one often writes $g(r) = \log r$.

Example 7.3 (Hard density cap). For $\kappa > 0$, define

$$G(r) = \iota_{[0, \kappa]}(r) = \begin{cases} 0, & 0 \leq r \leq \kappa, \\ +\infty, & \text{otherwise.} \end{cases}$$

This is convex but nonsmooth. Its subgradient is a pressure that is inactive below the cap and active where $\rho = \kappa$.

Example 7.4 (A nonconvex choice). For $G(r) = -r^2/2$, the kinetic term remains convex after the momentum change, but the full problem does not. Multiple local minima and concentration instabilities can occur. The momentum substitution cannot remove nonconvexity already present in G .

8 Benamou–Brenier, JKO, and variational MFG

8.1 Benamou–Brenier

Quadratic dynamic optimal transport is

$$\frac{1}{2} \mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\rho, m} \left\{ \int_0^1 \int J(\rho_t, m_t) dx dt : \partial_t \rho + \operatorname{div} m = 0, \rho_{t=0} = \rho_0, \rho_{t=1} = \rho_1 \right\}. \quad (13)$$

The variational MFG adds the running potential $\int_0^1 \int G(\rho_t)$ and leaves the endpoint free, selecting it through $\int \Psi \rho_1$. It is thus a convex dynamic transport problem, but not a distance between two prescribed measures. Setting $G = 0$ and fixing the endpoint recovers Benamou–Brenier (Benamou and Brenier, 2000).

8.2 JKO

A JKO step for an energy f is

$$\alpha^{k+1} \in \arg \min_{\alpha} \left\{ \frac{1}{2\tau} \mathcal{W}_2^2(\alpha^k, \alpha) + f(\alpha) \right\}. \quad (14)$$

Using (13), it is a one-period planning problem with a free endpoint and terminal functional $f(\alpha_1)$. It has no running congestion unless one adds it explicitly.

A variational MFG minimizes one full-horizon action containing $\int_0^1 G(\rho_t) dt$ and encodes anticipation over that horizon. A JKO scheme repeatedly solves short proximal problems; its small-step limit is a dissipative Wasserstein gradient flow (Jordan et al., 1998; Ambrosio et al., 2008).

Remark 8.1 (The common core). Benamou–Brenier is endpoint-to-endpoint transport, JKO is repeated short-horizon descent, and a potential MFG is a full-horizon variational game. All three share the convex mass–momentum perspective.

9 Diffusion and discretization

Diffusion. With idiosyncratic Brownian noise of variance 2ν , use

$$\partial_t \rho + \operatorname{div} m = \nu \Delta \rho.$$

This is still affine in (ρ, m) , so convexity is unchanged. The optimality system becomes

$$-\partial_t u - \nu \Delta u + \frac{1}{2} |\nabla u|^2 = G'(\rho), \quad \partial_t \rho - \nu \Delta \rho - \operatorname{div}(\rho \nabla u) = 0.$$

For quadratic drift costs, the problem also has an entropy-minimization interpretation on path space and admits Sinkhorn-like algorithms (Benamou et al., 2019).

Discrete convex program. After conservative discretization, collect cell densities in r and edge fluxes in M . The continuity equation becomes the affine system

$$Ar + BM = b.$$

The objective is a sum of local convex perspective and congestion terms plus a terminal linear term. Hence the discretized problem is finite-dimensional and convex. Primal–dual splitting, ADMM, and augmented-Lagrangian methods exploit the split between local proximal operations and the global linear conservation constraint (Benamou and Carlier, 2015; Achdou and Laurière, 2020).

10 Summary

The precise answer is

the momentum formulation is convex whenever G is convex.

The reason is not that the original density–velocity problem was secretly jointly convex. Rather:

1. $m = \rho v$ makes conservation affine;
2. the kinetic term becomes the closed perspective $J(\rho, m) = |m|^2/(2\rho)$;
3. this perspective is jointly convex and lower semicontinuous, including at vacuum;
4. convex congestion and convex endpoint penalties preserve convexity.

If G is nonconvex, the full problem remains nonconvex. When G is convex, the KKT equations are exactly the Hamilton–Jacobi/continuity system of a potential MFG, with coupling $g = G'$.

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